Appendix C

Full Proofs

This appendix is the authoritative proof surface for the defended ORIUS theorem program, but it is not a flat headline ladder. The active promotion gate now distinguishes flagship theorems, flagship corollaries, flagship lemmas, and archived-only historical text. No draft theorem row remains in the defended theorem register. T9, T10, `T11_Byzantine`, `T_stale_decay`, `T_minimax`, and `T_sensor_converse` are promoted only in their scoped forms, with domain-discharge artifacts binding the mathematical contracts to Battery, AV, and Healthcare evidence. The battery witness remains the deepest proof chain, so the full step-by-step derivations below begin with the reference ladder:

1. OASG existence,
2. one-step safety preservation,
3. the ORIUS core bound,
4. observation necessity / no free safety,
5. certificate validity horizon,
6. certificate expiration bound,
7. feasible fallback existence, and
8. graceful degradation dominance.

The certificate half-life result remains a corollary, and safe-budget monotonicity is retained as a narrow runtime law rather than a broad converse. The broader integrated theorem-like surface, including precursor results and supporting chapter statements, is catalogued separately in Appendix **??**. The active promotion truth source is `reports/publication/theorem_promotion_matrix.csv`.

Mathematical contract vs. domain discharge. The proof copies below use only stated assumptions and typed obligations as mathematical premises. Domain discharge is evidence, not an extra hidden assumption: the discharge artifacts instantiate constants, verify boundary-mass and mixing witnesses, and bind the proof kernels to the promoted bounded Battery, AV, and Healthcare artifacts. Thus T9/T10/T11 are contract-universal, not unrestricted-global; new physical-AI domains may reuse the proof pattern only after their adapters, safe sets, coverage witnesses, and fallback obligations are discharged by domain-specific evidence.

C.1 C.1 Battery-Domain OASG Existence

Theorem C.1 (restated)**.** *Under A1, A2, A4, A11, and A12, for dropout rate $d > 0$ and any optimal arbitrage controller, $\exists\, t^*$ with $\phi_{\text{obs}}(a_{t^*}) = 1$, $\phi_{\text{true}}(a_{t^*}) = 0$.*

Proof. Fix $\eta > 0$. By A11, the battery witness controller can be driven to a boundary-sensitive state under admissible demand, so there exists a step t^* and an admissible released action a_{t^*} such that the observed-state forward prediction lies on the safe side of the upper SOC boundary while remaining arbitrarily close to it:

$$\text{SOC}_{\max} - \eta \leq \hat{x}_{t^*} + \Delta(a_{t^*}) \leq \text{SOC}_{\max}.$$

By A2 and the standing dropout rate $d > 0$, there exists an admissible positive observation gap δ on a degraded-telemetry window. Choose that gap so that $\eta < \delta \leq g_{\max}$ and let

$$x_{t^*} = \hat{x}_{t^*} + \delta.$$

By A12, the fault process is conditionally independent of the controller action stream given the history, so the boundary-sensitive witness step and the admissible telemetry fault can co-occur on the same realized history. Now choose the admissible one-step model residual $\epsilon_{t^*} = 0$, which is allowed by A1 because $0 \in [-\epsilon_{\text{model}}, \epsilon_{\text{model}}]$.

The controller evaluates the action through the observed state only, so the observed successor satisfies

$$\hat{x}_{t^*+1} = \hat{x}_{t^*} + \Delta(a_{t^*}) \leq \text{SOC}_{\max},$$

hence $\phi_{\text{obs}}(a_{t^*}) = 1$. Using A4 for the shared one-step dynamics, the true successor is

$$x_{t^*+1} = x_{t^*} + \Delta(a_{t^*}) + \epsilon_{t^*} = \hat{x}_{t^*} + \Delta(a_{t^*}) + \delta.$$

Combining this identity with the lower side of the observed-state witness gives

$$x_{t^*+1} \geq (\mathrm{SOC}_{\max} - \eta) + \delta = \mathrm{SOC}_{\max} + (\delta - \eta) > \mathrm{SOC}_{\max}.$$

Therefore $\phi_{\mathrm{true}}(a_{t^*}) = 0$ while $\phi_{\mathrm{obs}}(a_{t^*}) = 1$, which is exactly the battery witness-row observation-action safety gap. This is an existence witness, not a frequency or universality claim: A11 supplies the boundary-sensitive step, A12 permits the fault/controller co-occurrence, and a new domain cannot reuse T1 unless it discharges the same boundary-reachability obligation. □

C.2 C.2 One-Step Safety Preservation

Theorem C.2 (restated)**.** *Under A1–A5, A7, if $a_t^{\mathrm{safe}} \in \mathcal{A}_t$ and $x_t \in \hat{C}_t^{\mathrm{RAC}}$, then $\phi(x_{t+1}) = 1$, where the tightened margin used to define $\mathcal{A}_t$ already absorbs the one-step model-error allowance.*

Proof. Fix a step t and assume the standing conditions A1–A5 and A7. Because $x_t \in \hat{C}_t^{\mathrm{RAC}}$, the true state lies in the reliability-adjusted conformal tube issued from the current observation. In particular, membership in that tube gives

$$|x_t - \hat{x}_t| \leq m_t,$$

where m_t is the inflation-adjusted half-width determined by A2 and A5. This step is deterministic once the theorem is conditioned on $x_t \in \hat{C}_t^{\mathrm{RAC}}$; it does not introduce a new probability statement. By A5, the shield uses the absorbed tightening margin

$$m_t^\star = m_t + \epsilon_{\mathrm{model}}$$

when constructing $\mathcal{A}_t$. Since $a_t^{\mathrm{safe}} \in \mathcal{A}_t$, the observed-state forward prediction therefore satisfies

$$\mathrm{SOC}_{\min} + m_t^\star \leq \hat{x}_t + \Delta(a_t^{\mathrm{safe}}) \leq \mathrm{SOC}_{\max} - m_t^\star,$$

with $\Delta(a_t^{\mathrm{safe}})$ denoting the one-step battery state update implied by the known dynamics from A4. Now write the true next state as

$$x_{t+1} = x_t + \Delta(a_t^{\mathrm{safe}}) + \epsilon_t,$$

where the model residual ϵ_t is bounded by A1. Substituting $x_t = \hat{x}_t + (x_t - \hat{x}_t)$ and using

$|x_t - \hat{x}_t| \le m_t$ yields

$$x_{t+1} \ge (\hat{x}_t + \Delta(a_t^{\text{safe}})) - m_t - |\epsilon_t| \ge \text{SOC}_{\min} + m_t^\star - m_t - \epsilon_{\text{model}} = \text{SOC}_{\min} + (m_t + \epsilon_{\text{model}}) - m_t - \epsilon_{\text{model}} = \text{SOC}_{\min}.$$

The same substitution on the upper side gives

$$x_{t+1} \le (\hat{x}_t + \Delta(a_t^{\text{safe}})) + m_t + |\epsilon_t| \le \text{SOC}_{\max} - m_t^\star + m_t + \epsilon_{\text{model}} = \text{SOC}_{\max} - (m_t + \epsilon_{\text{model}}) + m_t + \epsilon_{\text{model}} = \text{SOC}_{\max}.$$

Therefore

$$\text{SOC}_{\min} \le x_{t+1} \le \text{SOC}_{\max}.$$

This is exactly the safety predicate $\phi(x_{t+1}) = 1$. The argument uses only the current-step information certified by A7 and therefore preserves causality. Hence any action selected from $\mathcal{A}_t$ while the current state is inside $\hat{C}_t^{\text{RAC}}$ keeps the one-step successor in the safe set. The theorem stops at this one-step conditional statement; it does not itself quantify how often $x_t \in \hat{C}_t^{\text{RAC}}$ occurs. □

C.3 C.3 Battery ORIUS Core Bound

Theorem C.3 (restated)**.** *Under A1–A5, A7, and the explicit predictable battery risk-envelope contract* $\mathbb{P}[Z_t = 1 \mid \mathcal{H}_t] \le r_t$ *with* $r_t = \alpha(1 - w_t)$ *on the fixed certified budget trace, one has* $\mathbb{E}[V] \le \alpha\, T\, (1 - \bar{w})$ *for* $\bar{w} = T^{-1} \sum_{t=1}^{T} w_t$.

Proof. Let

$$V = \sum_{t=1}^{T} Z_t, \qquad Z_t = \mathbf{1}\big[\neg\phi(x_t)\big],$$

so that V counts the number of true-state safety violations across the episode. Because each Z_t is an indicator, $\mathbb{E}[Z_t] = \mathbb{P}[Z_t = 1]$. Applying linearity of expectation, which does not require independence across time, gives

$$\mathbb{E}[V] = \sum_{t=1}^{T} \mathbb{E}[Z_t] = \sum_{t=1}^{T} \mathbb{P}[Z_t = 1].$$

The theorem-local battery contract supplies the per-step conditional envelope

$$\mathbb{P}[Z_t = 1 \mid \mathcal{H}_t] \le r_t, \qquad r_t = \alpha(1 - w_t),$$

where r_t is predictable from the runtime history used to issue the certificate. In the fixed replay/certificate trace used by the theorem artifact, the recorded budget values are treated

as constants. If the reliability process is left random instead, the defensible statement is the weaker averaged form $\mathbb{E}[V] \le \alpha \sum_t \mathbb{E}[1 - w_t]$. Taking expectations of both sides gives

$$\mathbb{P}[Z_t = 1] = \mathbb{E}[\mathbb{P}[Z_t = 1 \mid \mathcal{H}_t]] \le \mathbb{E}[r_t].$$

On the fixed recorded budget trace, $\mathbb{E}[r_t] = r_t$, giving the next display.

Substituting the per-step bound into the expectation identity yields

$$\mathbb{E}[V] \le \sum_{t=1}^{T} r_t = \sum_{t=1}^{T} \alpha(1 - w_t) = \alpha \sum_{t=1}^{T} (1 - w_t).$$

Now expand the sum:

$$\sum_{t=1}^{T} (1 - w_t) = T - \sum_{t=1}^{T} w_t = T \left(1 - \frac{1}{T} \sum_{t=1}^{T} w_t\right).$$

By definition of the episode-average reliability,

$$\bar{w} = \frac{1}{T} \sum_{t=1}^{T} w_t.$$

Replacing the bracketed term with $\bar{w}$ gives

$$\mathbb{E}[V] \le \alpha \, T \, (1 - \bar{w}).$$

This is the desired aggregation corollary. The actual burden is the per-step envelope hypothesis $\mathbb{P}[Z_t = 1 \mid \mathcal{H}_t] \le \alpha(1 - w_t)$; once that T3a-style derivation is available, the displayed $\mathbb{E}[V] \le \alpha T(1 - \bar{w})$ bound follows by linearity of expectation alone. The derivation invokes only finite-horizon identities and the explicit theorem-local battery risk budget; no temporal independence, exchangeability across time, or martingale strengthening is introduced. Therefore the battery ORIUS core bound holds for any admissible horizon T on the defended replay surface whenever that per-step budget contract has been justified. □

C.4 C.4 Observation Necessity / No Free Safety

Theorem C.4 (restated)**.** *Under A1, A2, A4, and A11, let π be a fixed-margin, quality-ignorant mandatory-release controller with effective margin m^π. If the battery witness row has an admissible degraded-observation gap $\delta > m^\pi + \eta$ at a boundary-sensitive release step for some $\eta > 0$, then there exists an admissible fault sequence with* $\mathrm{TSVR}(\pi) > 0$.

Proof. Fix such a controller π. By definition, its effective safety margin does not depend on the runtime reliability score, so there exists a constant margin $m^\pi \geq 0$ used across all observation qualities. By A11, the battery witness controller can be driven to a boundary-sensitive state under admissible demand. Consequently, for every $\eta > 0$ there exists a step $t^\star$ and an action released by π such that the observed-state forward prediction nearly saturates the controller's fixed margin on the upper SOC boundary:

$$\hat{x}_{t^\star} + \Delta(\pi(\hat{x}_{t^\star})) \geq \mathrm{SOC}_{\max} - m^\pi - \eta.$$

By the theorem's gap hypothesis and A2, choose a degraded-observation fault realization with

$$\delta \in (m^\pi + \eta,\ g_{\max}], \qquad x_{t^\star} = \hat{x}_{t^\star} + \delta.$$

Now set the admissible one-step model residual to zero, which is allowed by A1. Using A4 for the shared battery dynamics,

$$x_{t^\star+1} = x_{t^\star} + \Delta(\pi(\hat{x}_{t^\star})) + \epsilon_{t^\star} = \hat{x}_{t^\star} + \Delta(\pi(\hat{x}_{t^\star})) + \delta.$$

Substituting the observed-state witness inequality gives

$$x_{t^\star+1} \geq \mathrm{SOC}_{\max} - m^\pi - \eta + \delta = \mathrm{SOC}_{\max} + (\delta - m^\pi - \eta) > \mathrm{SOC}_{\max}.$$

The observed state therefore still appears feasible relative to the fixed margin m^π, while the same released action produces a true-state boundary crossing. Hence the controller incurs at least one true-state violation on the constructed admissible sequence, so $\mathrm{TSVR}(\pi) > 0$. The witness is finite-horizon and class-scoped exactly as claimed in T4. It does not rule out controllers with adaptive margins, abstention, fallback, or a fixed margin so large that the stated degraded-observation gap is impossible. □

C.5 C.5 Certificate Validity Horizon

Proof source. This section restates T5, *Finite-Horizon Certificate Validity*.

Theorem T5: Finite-Horizon Certificate Validity. Let X_t be the observation-consistent state set at issue time t, let $\tilde{a}_{t:t+h}$ be the certified action or fallback sequence, and let S be the domain safe-state set. Assume the forward reachable tube satisfies

$$\mathcal{R}_{t+j}(X_t, \tilde{a}_{t:t+j}) \subseteq S \qquad j = 0, \ldots, h.$$

Then every release governed by the certificate is safe for h steps unless a certificate-invalidating event occurs.

Proof. At $j = 0$, all states consistent with the issued certificate lie in $\mathcal{R}_t(X_t) \subseteq S$. Suppose the claim holds through step $j < h$. By definition of the reachable tube, every state reachable after applying the certified prefix $\tilde{a}_{t:t+j+1}$ lies in $\mathcal{R}_{t+j+1}(X_t, \tilde{a}_{t:t+j+1})$. The containment assumption places that set inside S. Induction proves safety for all $j = 0, \ldots, h$. If a fresh observation contradicts X_t, the model, or the action sequence, the certificate is outside the theorem hypothesis and must be renewed, repaired, or expired by the runtime.

C.6 C.6 Certificate Expiration Bound

Theorem C.5 (restated)**.** *Under A4, A7, A9 and confidence level $\delta \in (0, 1)$, the battery certificate horizon satisfies the lower bound*

$$\tau_t \geq \left\lfloor \frac{\delta_{\mathrm{bnd},t}^2}{2\sigma_d^2 \log(2/\delta)} \right\rfloor .$$

Proof. For a σ_d-sub-Gaussian disturbance, the reflection-principle / first-passage tail inequality gives

$$\mathbb{P}\Big(|S_\Delta| \geq \delta_{\mathrm{bnd},t}\Big) \leq 2 \exp\left(-\frac{\delta_{\mathrm{bnd},t}^2}{2\sigma_d^2 \Delta}\right).$$

Demanding the right-hand side be at most δ and solving for Δ yields

$$\Delta \leq \frac{\delta_{\mathrm{bnd},t}^2}{2\sigma_d^2 \log(2/\delta)}.$$

Flooring gives the integer lower bound, and the runtime helper `certificate_expiration_bound()` implements this confidence-aware expression directly. □

C.7 C.7 Certificate Half-Life Corollary

Corollary C.6 (restated)**.** *The certificate half-life remains a corollary rather than part of the flagship T1–T8 ladder.*

Proof. The current battery blackout evidence supports a bounded safe-hold claim more strongly than a finite empirically identified half-life law. The corollary therefore remains analytical and should be interpreted together with the more conservative wording in Chapter **??**. □

C.8 C.8 Feasible Fallback Existence

Theorem C.7 (restated)**.** *Under A1, A4, and A8, there exists a battery piecewise fallback rule that either safely holds on the interior or executes a boundary-aware safe-landing recovery action; otherwise the runtime fails closed instead of asserting a false fallback guarantee.*

Proof. Zero dispatch is the constructive hold action on the interior. If the current SOC has enough slack against the one-step model error, the next state remains inside the admissible interval. If the current SOC is safe but too close to a boundary for zero dispatch to be certified, the safe-landing controller computes a charge or discharge action that moves the SOC toward the interior safe zone while respecting the active battery limits. The helper `certify_fallback_existence()` now checks exactly this piecewise surface and returns infeasible rather than over-claiming when neither case is safe. □

C.9 C.9 Graceful Degradation Dominance

Proof source. This section restates T8, *Graceful Degradation Dominance with Useful Work*.

Theorem T8: Graceful Degradation Dominance with Useful Work. Fix a finite horizon and a paired admissible fault trace. Let P_G be an ORIUS graceful-degradation policy and P_U an uncontrolled persistence policy evaluated on that same trace. Let $Z_t^G, Z_t^U \in \{0,1\}$ be true-state violation indicators, and let $W_G, W_U \geq 0$ be the corresponding useful-work totals on the horizon. For a declared operating threshold $\lambda \in (0,1]$, if

$$Z_t^G \leq Z_t^U \quad \forall t \qquad \text{and} \qquad W_G \geq \lambda W_U,$$

then P_G weakly dominates P_U in the safety–work preorder

$$P \preceq_\lambda Q \quad \Longleftrightarrow \quad \sum_t Z_t^P \leq \sum_t Z_t^Q \ \text{ and } \ W_P \geq \lambda W_Q.$$

This is a paired-trace dominance statement, not a claim that P_G is globally optimal among all fallback policies.

Proof. Summing the pointwise inequality over the fixed horizon gives

$$\sum_t Z_t^G \leq \sum_t Z_t^U,$$

so the graceful policy has no larger violation count, and therefore no larger violation rate, on the paired trace. The second hypothesis is exactly the useful-work lower bound $W_G \geq \lambda W_U$. These two inequalities are the two coordinates of the stated preorder. Immediate shutdown can satisfy the safety coordinate, but it is a T8 witness only if it also satisfies the same useful-work threshold; otherwise it is outside the theorem's dominance claim.

C.10 C.10 Safe-Budget Monotonicity Proposition

Proposition C.8 (restated)**.** *Under A1–A5, $B_t = |\mathcal{A}_t| \cdot \Delta t$ is monotone increasing in w_t.*

Proof. $|\mathcal{A}_t| = \mathrm{SOC}_{\max} - \mathrm{SOC}_{\min} - 2m_t$ and $m_t = q_t/(w_t + \epsilon)$ is monotone decreasing in w_t, so B_t increases with w_t. □

C.11 C.11 Impossibility of Quality-Ignorant Mandatory Release (T9)

Theorem T9: Impossibility of Quality-Ignorant Mandatory Release. Let Π_{obs} be total observation-only mandatory-release controllers $\pi : O \to A$, with no abstention or fail-closed action in the policy class. Assume $C(x) \subseteq A$ is the state-conditioned safe-action set. For an observation o with nonempty ambiguity class, define

$$B(o) = \{x \in X : h(x) = o\}, \qquad K(o) = \bigcap_{x \in B(o)} C(x).$$

If $B(o) \neq \varnothing$ and $K(o) = \varnothing$, no controller in Π_{obs} can guarantee true-state safety for all $x \in B(o)$.

Proof. A total mandatory observation-only controller must choose one action $a = \pi(o)$ for the entire nonempty ambiguity class $B(o)$. If that action were safe for every $x \in B(o)$, then

$$a \in \bigcap_{x \in B(o)} C(x) = K(o),$$

contradicting $K(o) = \varnothing$. Hence there exists at least one $x^\dagger \in B(o)$ with $a \notin C(x^\dagger)$, so the release cannot be certified true-state safe on the whole ambiguity class. The claim is restricted to mandatory release; abstention, fallback, uncertainty expansion, or denial of release are outside Π_{obs}.

C.12 C.12 Boundary-Indistinguishability Lower Bound (T10)

Theorem T10: Boundary-Indistinguishability Lower Bound. Let x_0, x_1 be two latent boundary states with observation laws P_0, P_1. Assume $d_{\mathrm{TV}}(P_0, P_1) \le \epsilon$ and $C(x_0) \cap C(x_1) = \varnothing$. For any observation-only mandatory-release controller π, deterministic or randomized,

$$\sup_{i \in \{0,1\}} \Pr_{O \sim P_i}[\pi(O) \notin C(x_i)] \ge \frac{1-\epsilon}{2}.$$

Proof. It suffices to prove the deterministic case; randomized controllers follow by conditioning on the controller's internal random seed and averaging. Define

$$S_i = \{o : \pi(o) \in C(x_i)\}, \qquad i \in \{0,1\}.$$

The disjoint-safe-set hypothesis gives $S_0 \cap S_1 = \varnothing$. The average success probability of the release rule is

$$\frac{1}{2}\{P_0(S_0) + P_1(S_1)\}.$$

Since $S_1 \subseteq S_0^c$, the binary test $\delta(o) = 0$ for $o \in S_0$ and $\delta(o) = 1$ otherwise has testing success

$$\frac{1}{2}\{P_0(S_0) + P_1(S_0^c)\} \ge \frac{1}{2}\{P_0(S_0) + P_1(S_1)\}.$$

The best possible two-hypothesis testing success between P_0 and P_1 is $(1 + d_{\mathrm{TV}}(P_0, P_1))/2$. Therefore the release rule's average success is at most $(1+\epsilon)/2$, so its average loss is at least $(1-\epsilon)/2$. The larger of the two state-conditioned release risks is at least that average loss, which proves the bound.

C.13 C.13 Typed Structural Transfer and Failure-Mode Converse (T11)

Theorem reference. The forward theorem is Theorem **??** and the converse witness is Proposition **??** in Chapter **??**. The appendix copies the key proof logic so the full-proofs surface remains self-contained.

Appendix proof copy. Fix a step t and condition on the causal observation history $\mathcal{H}_t$. Let $E_t = \{x_t \in U_t\}$ denote the event that the latent state lies in the constructed observation-

consistent set. By the first T11 obligation,

$$\mathbb{P}[E_t \mid \mathcal{H}_t] \geq 1 - \alpha.$$

On E_t, the soundness obligation implies that every action in the tightened safe-action set preserves safety at the next step for every state in U_t and every admissible disturbance. If the tightened safe-action set is non-empty, the repair-membership obligation guarantees that the released repaired action lies inside that set and is therefore safe on E_t. If the tightened safe-action set is empty, the fallback-admissibility obligation guarantees a safe release path from the current safe state. In either branch,

$$\mathbb{P}[x_{t+1} \in \mathcal{S} \mid \mathcal{H}_t] \geq \mathbb{P}[E_t \mid \mathcal{H}_t] \geq 1 - \alpha.$$

This proves the forward one-step transfer direction. The converse is not claimed as part of T11 itself; it is tracked separately as a structural failure-mode witness showing that if any one of the four obligations fails, the corresponding step in the forward proof disappears. □

Corollary C.9 (T10_T11_ObservationAmbiguitySandwich: Covered Observation-Ambiguity Safety Sandwich)**.** *Let $B(o) = \{x : h(x) = o\}$ denote an observation ambiguity class and let $K(o) = \bigcap_{x \in B(o)} C(x)$ denote its common safe core. Every mandatory-release observation-only controller satisfies*

$$\mathrm{TSVR}(\pi) \geq \mathbb{E}_O\left[\min_{a \in \mathcal{A}} \mathbb{P}(a \notin C(X^\star) \mid O)\right].$$

If ORIUS constructs a set $\mathcal{X}_t$ with $\mathbb{P}(X_t^\star \notin \mathcal{X}_t) \leq \alpha$ and releases only $u_t \in \bigcap_{x \in \mathcal{X}_t} C(x)$, then $\mathrm{TSVR}_{\mathrm{ORIUS}} \leq \alpha$; under exact coverage, the one-step true-state violation probability is zero.

Proof. Conditioning on $O = o$, an observation-only controller must select one action for all states in $B(o)$. Its conditional violation probability is $\mathbb{P}(a \notin C(X^\star) \mid O = o)$, so no such controller can improve on the minimum over a. Averaging over observations gives the Bayes lower bound. When $K(o) = \emptyset$, no action is safe for all states in the ambiguity class, so any mandatory release has positive conditional risk if the unsafe state has positive conditional mass.

For ORIUS, on the event $X_t^\star \in \mathcal{X}_t$, membership of u_t in every $C(x)$ for $x \in \mathcal{X}_t$ implies $u_t \in C(X_t^\star)$, hence no true-state violation. The remaining violation probability is bounded by the coverage miss probability α. □

Lemma C.10 (T11_AV_BrakeHold: AV Brake-Hold Runtime Lemma)**.** *For a promoted ORIUS autonomous-vehicle replay row, suppose the emitted domain runtime contract witness has source theorem T11, T11 status* ***runtime_linked****, no failed T11 obligations, a valid*

certificate, and a passing true brake-hold postcondition in the runtime trace. Then that row is a bounded runtime instance of the forward T11 transfer surface for the longitudinal brake-hold contract.

Proof. The witness records the four executable T11 obligations through the typed contract summary and records the AV-specific true postcondition separately as `domain_postcondition_passed`. The stated hypotheses therefore instantiate the coverage, soundness, repair-membership, and fallback admissibility steps used in the T11 proof, and the runtime postcondition checks the narrowed brake-hold safety predicate for the emitted next state. The result is only row-local and contract-local; it does not assert complete autonomous-driving closure. □

Lemma C.11 (T11_HC_FailSafeRelease: Healthcare Fail-Safe Release Runtime Lemma)**.** *For a promoted ORIUS healthcare monitoring row, suppose the emitted domain runtime contract witness has source theorem T11, T11 status* ***runtime_linked****, no failed T11 obligations, a valid certificate, and a passing true fail-safe alert-release postcondition in the runtime trace. Then that row is a bounded runtime instance of the forward T11 transfer surface for the healthcare fail-safe release contract.*

Proof. The witness records the typed T11 summary emitted by the universal runtime and separately records the healthcare-specific true alert-release predicate. If the T11 status is not `runtime_linked`, if any obligation fails, if the certificate is invalid, or if the postcondition fails, the witness is not passing. A passing witness therefore supplies exactly the forward one-step T11 obligations plus the narrowed clinical-monitoring runtime postcondition. It does not assert regulated clinical deployment closure. □

Lemma C.12 (T6_AV_FallbackValidity: AV One-Step Fallback Certificate Lemma)**.** *For a promoted ORIUS AV replay row, if the certificate records the* ***single_step_fallback*** *validity scope, the released action is the full-brake fail-safe action, the T11 runtime witness is linked, and the true brake-hold postcondition passes, then the certificate has a theorem-backed one-step validity horizon for that row.*

Proof. The helper implementing the certificate semantics assigns $H_t = 1$, half-life one, and expiration at the next step only in the fallback branch where the emitted action is the domain fail-safe action. The runtime witness separately checks T11 linkage and the true brake-hold postcondition. Thus the certificate is valid only for the single emitted transition; no multi-step validity is inferred from degraded observation. □

Lemma C.13 (T6_HC_FallbackValidity: Healthcare One-Step Fallback Certificate Lemma)**.** *For a promoted ORIUS healthcare monitoring row, if the certificate records the* ***single_step_fallback***

validity scope, the released action is the max-alert fail-safe action, the T11 runtime witness is linked, and the true fail-safe alert-release postcondition passes, then the certificate has a theorem-backed one-step validity horizon for that row.

Proof. The certificate-validity helper gives fallback releases a positive horizon only for the domain fail-safe action and only for one step. T11 linkage and the healthcare postcondition are checked by the runtime witness. Therefore the row satisfies the formal validity predicate for the emitted transition, while degraded observations are not promoted to multi-step certificates. □

Lemma C.14 (T_EQ_Battery_RuntimeArtifactPackage: Battery Equal Artifact Discipline Package)**.** *If the battery report directory contains runtime-denominator comparator, ablation, negative-control, trace, utility, theorem, proof-appendix, and reproducibility artifacts satisfying the equal-domain gate, then the battery row satisfies the repository's equal artifact-discipline contract.*

Proof. The equal-domain gate is an executable artifact obligation, not a new safety theorem. It checks that the battery evidence package contains a runtime trace, runtime-native comparator rows, runtime-native ablation rows, runtime-native negative controls, a proof-surface anchor, and a reproducibility manifest. It also checks the non-vacuity condition that the ORIUS row has zero TSVR, certificate/witness closure, and useful work greater than the degenerate safe-fallback comparator. If any required artifact is missing or proxy-labelled, the validator fails closed. Therefore a passing gate is exactly the stated artifact-discipline contract. □

Lemma C.15 (T_EQ_AV_RuntimeArtifactPackage: AV Equal Artifact Discipline Package)**.** *If the AV report directory contains runtime-denominator comparator, ablation, negative-control, trace, utility, theorem, proof-appendix, and reproducibility artifacts satisfying the equal-domain gate, then the bounded AV replay row satisfies the repository's equal artifact-discipline contract.*

Proof. The gate requires the AV T11 brake-hold runtime lemma and the AV T6 one-step fallback-validity lemma, then separately requires domain-native runtime denominator rows for baselines, ablations, negative controls, utility, and reproducibility. The ORIUS row must have zero TSVR, valid runtime certificates, passing runtime witnesses, and useful work greater than the always-brake comparator. The conclusion is limited to artifact discipline for the promoted replay contract; it does not assert full autonomous-driving closure. □

Lemma C.16 (T_EQ_HC_RuntimeArtifactPackage: Healthcare Equal Artifact Discipline Package)**.** *If the healthcare report directory contains runtime-denominator comparator, ablation, negative-control, trace, utility, theorem, proof-appendix, and reproducibility artifacts*

satisfying the equal-domain gate, then the bounded healthcare monitoring row satisfies the repository's equal artifact-discipline contract.

Proof. The gate requires the healthcare T11 fail-safe release runtime lemma and the healthcare T6 one-step fallback-validity lemma, then requires native runtime denominator rows for baselines, ablations, negative controls, utility, and reproducibility. The ORIUS row must have zero TSVR, valid runtime certificates, passing runtime witnesses, and useful work greater than the always-alert comparator. The result is an artifact-discipline guarantee for the promoted monitoring contract only; it does not assert regulated clinical deployment readiness. □

C.14 L1 — Reliability-Monotone Inflation

Lemma L1: Reliability-Monotone Inflation. Let $m_t = q_t/(w_t + \epsilon)$ with $q_t > 0$, $w_t \in [0, 1]$, and $\epsilon > 0$. Then m_t is monotonically nonincreasing in w_t.

Proof.

$$\frac{\partial m_t}{\partial w_t} = -\frac{q_t}{(w_t + \epsilon)^2} < 0.$$

Thus increasing reliability cannot increase the margin.

C.15 L2 — Safe-Set Antitonicity

Lemma L2: Safe-Set Antitonicity. If $X_1 \subseteq X_2$, then

$$\bigcap_{x \in X_2} C(x) \subseteq \bigcap_{x \in X_1} C(x).$$

Proof. Any action safe for every state in the larger set X_2 is safe for every state in its subset X_1. Hence the larger uncertainty set has a smaller or equal common safe-action set.

C.16 L3 — Intervention Threshold

Lemma L3: Intervention Threshold. If a candidate action a_t lies outside $\cap_{x \in X_t} C(x)$, any certified ORIUS release must repair, fallback, or deny release.

Proof. Membership in $\cap_{x \in X_t} C(x)$ is exactly the condition for the action to be safe for every state covered by the runtime uncertainty set. If the action is outside that set, releasing it

would violate the covered-release contract. Therefore a certified runtime must choose a repair, fallback, or denial path.

C.17 L4 — Observation-Ambiguity Safety Sandwich

Lemma L4: Observation-Ambiguity Safety Sandwich. Observation-only mandatory release inherits the lower-side ambiguity risks in T9 and T10. ORIUS covered release has upper-side risk bounded by its coverage miss probability under the T2/T3 covered-release contract.

Proof. The lower side follows by applying T9 to empty safe cores and T10 to two-state boundary ambiguity classes. For the upper side, ORIUS releases only actions safe over the covered state set. Thus violation can occur only when the true state is outside that covered set or an explicit runtime assumption is broken. Under the T2/T3 contract this event has probability no larger than the declared coverage miss probability. Combining the lower and upper statements gives the sandwich.

C.18 T11_Byzantine — Byzantine-Robust Reliability Aggregation

Theorem `T11_Byzantine`: Byzantine-Robust Reliability Aggregation. Suppose n telemetry channels emit scores in $[0,1]$, at most $b < n/2$ are Byzantine, and $\hat{w}_t$ is the b-trimmed mean after sorting all reported scores and removing the b smallest and b largest values. If the honest scores lie in an interval of radius ρ around w_t^H, then

$$|\hat{w}_t - w_t^H| \leq \rho.$$

Proof. Trimming removes the b smallest and b largest reported values. Since at most b reports are Byzantine, no value below all honest scores can remain after the lower trim: there are at most b such values, so they are among the b smallest reports. Similarly, no value above all honest scores can remain after the upper trim. All honest scores lie in $[w_t^H - \rho, w_t^H + \rho]$ by hypothesis. Hence every retained value that contributes to the trimmed mean lies in $[w_t^H - \rho, w_t^H + \rho]$. Their arithmetic mean lies in the same interval, which proves the bound. The theorem makes no claim for $b \geq n/2$ or for honest scores outside the stated interval.

C.19 T_stale_decay — Stale-Hold Uncertainty Growth

Theorem `T_stale_decay`: Stale-Hold Uncertainty Growth. Assume no fresh observation arrives for s steps and the latent state obeys $\|x^\star_{t+k} - x^\star_{t+k-1}\| \le L$ for $k = 1, \dots, s$. If the uncertainty radius at time t is r_t, then the conservative stale-hold radius is $r_{t+s} = r_t + Ls$.

Proof. By the triangle inequality,

$$\|x^\star_{t+s} - x^\star_t\| \le \sum_{k=1}^{s} \|x^\star_{t+k} - x^\star_{t+k-1}\| \le Ls.$$

Combining this drift radius with the existing uncertainty radius r_t gives $r_t + Ls$. Therefore any certified release at $t + s$ must be safe over the expanded set. This is a bounded-drift containment theorem, not a claim that a particular exponential decay schedule is physically derived.

C.20 T_minimax — Finite Ambiguity-Class Minimax Lower Bound

Theorem `T_minimax`: Finite Ambiguity-Class Minimax Lower Bound. Let $\mathcal{P}_\epsilon$ be a nonempty class of finite two-state boundary problems $P = (x_0, x_1, P_0, P_1, C)$ satisfying $d_{\mathrm{TV}}(P_0, P_1) \le \epsilon$ and $C(x_0) \cap C(x_1) = \varnothing$. For a mandatory observation-only controller π, define the worst-latent-state loss on P by

$$L(\pi, P) = \sup_{i \in \{0,1\}} \Pr_{O \sim P_i}[\pi(O) \notin C(x_i)].$$

Then

$$\inf_{\pi \in \Pi_{\mathrm{obs}}} \sup_{P \in \mathcal{P}_\epsilon} L(\pi, P) \ge \frac{1 - \epsilon}{2}.$$

Proof. Fix any π. For every $P \in \mathcal{P}_\epsilon$, the hypotheses of T10 hold by definition of the class, so

$$L(\pi, P) \ge \frac{1 - \epsilon}{2}.$$

Because the class is nonempty, taking the supremum over $P \in \mathcal{P}_\epsilon$ preserves the same lower bound for the fixed π. Taking the infimum over π then preserves it for the controller class. This is a finite ambiguity-class lower bound under the displayed loss; it is not a global minimax optimality theorem and does not assert a matching ORIUS upper bound.

C.21 T_sensor_converse — Sensor Necessity Under Adapter Semantics

Theorem `T_sensor_converse`: Sensor Necessity Under Adapter Semantics. Let x^j be a latent coordinate on which the safe-action map depends. Suppose the adapter observation map is invariant to that coordinate: whenever two states differ only in x^j, they have the same observation. If there exist x_0, x_1 differing only in x^j such that

$$h(x_0) = h(x_1) = o, \qquad C(x_0) \cap C(x_1) = \varnothing,$$

then no observation-only mandatory-release policy can certify safety over the ambiguity class containing x_0, x_1 without additional sensing, uncertainty expansion, fallback, or denial.

Proof. By the adapter-invariance hypothesis, x_0 and x_1 produce the same observation o, so both states lie in the nonempty ambiguity class $B(o)$. Because $C(x_0) \cap C(x_1) = \varnothing$, the common safe core over that class is empty. T9 then rules out certification by any total observation-only mandatory-release policy on $B(o)$. Additional sensing, uncertainty expansion, fallback, or denial are the escape routes because they change the observation map or leave the mandatory-release policy class.

C.22 T_trajectory_PAC — Finite-Horizon PAC Release Certificate

Theorem T_trajectory_PAC: Finite-Horizon PAC Release Certificate. Let Z_t be the true-state violation event at step t. Suppose

$$\Pr(Z_t = 1) \le \epsilon_t, \qquad t = 1, \dots, T.$$

Then

$$\Pr\left(\sum_{t=1}^{T} Z_t \ge 1\right) \le \sum_{t=1}^{T} \epsilon_t.$$

Therefore, if $\sum_{t=1}^{T} \epsilon_t \le \delta$, the trajectory is violation-free with probability at least $1 - \delta$. The same conclusion holds when the runtime supplies predictable conditional budgets $\Pr(Z_t = 1 \mid \mathcal{H}_{t-1}) \le \epsilon_t$, after taking expectations.

Proof. The event $\{\sum_{t=1}^{T} Z_t \geq 1\}$ is the union of the per-step violation events $\cup_{t=1}^{T}\{Z_t = 1\}$. By the union bound,

$$\Pr\left(\cup_{t=1}^{T}\{Z_t = 1\}\right) \leq \sum_{t=1}^{T} \Pr(Z_t = 1) \leq \sum_{t=1}^{T} \epsilon_t.$$

If the sum is at most δ, then the complementary no-violation event has probability at least $1 - \delta$. This defended surface is exactly the finite-horizon Bonferroni certificate; it does not claim a Ville or supermartingale strengthening and does not require temporal independence.